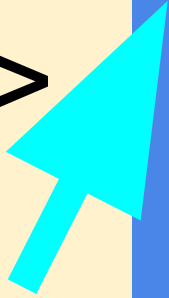


ONE PERSON IN GROUP-> CLICK USE TEMPLATE



WHEN THE COPIED TEMPLATE OF THESE SLIDES OPENS IN A NEW TAB:

- ADD YOUR Group NAME TO THE TITLE
- SHARE WITH GROUP MEMBERS (USE SHARE BUTTON)
- COPY THE TITLE AND LINK TO YOUR RECORDS (Digital folder/table of contents)
- Remember it is advisable to make a PERSONAL copy and link it when DONE

Solubility Table and Double Displacements

ESSENTIAL QUESTIONS

Remember, you should be able to thoroughly answer these AFTER learning through following resources.

- What are the predictable ways elements and compounds come together in double displacement reactions?
- How do precipitates form during double displacement reactions?
- How can we predict the solubility of various ionic compounds using a solubility table

Instructions

Follow through the **Engage to Apply** resources in order .

- Once complete, choose from Extend resources as interested.
- Record your learning in your **these slides- Remember to keep track of where to find these slides for studying!**
- Complete the **Essential Questions Response in the Apply Section** .
- Try the Extend section and optional Apply practice

Hyperdoc Elements

Click for quick access to:

- [Engage](#)
- [Explore](#)
- [Explain](#)
- [Apply](#)
- [Extend](#)

Terminology - As you encounter terms new to you, define in your own words here, include examples (make them up!)



Part 1: ENGAGE

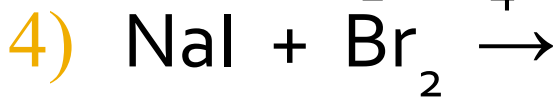
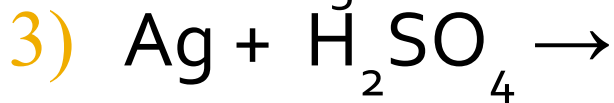
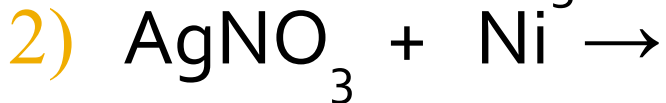
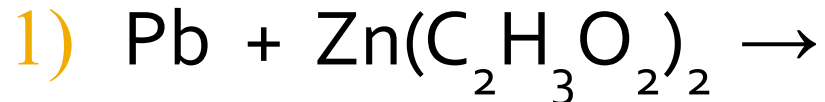
(5 mins)

As a group discuss the following

1. What do the terms soluble and insoluble mean?
2. Which of the following do you think are most likely to be insoluble? Hint: Recall what we learned about charge and size of ions.
3. LiCl MgS PbI_2

Speed challenge on SINGLE displacement reactions

Using your metal and halogen activity series, race against your group mates to predict whether the following reactions will take place. Click to the next slide to check your answers once you are done!



Speed challenge on SINGLE displacement reactions

Using your metal and halogen activity series, race against your group mates to predict whether the following reactions will take place. Click to the next slide to check your answers once you are done!

Move this box to see answers

Part 2: EXPLORE

Explore: Watch & learn

Pause at the 4 minute mark
and predict the
solubility of the
compounds.
Continue video
to check your
work

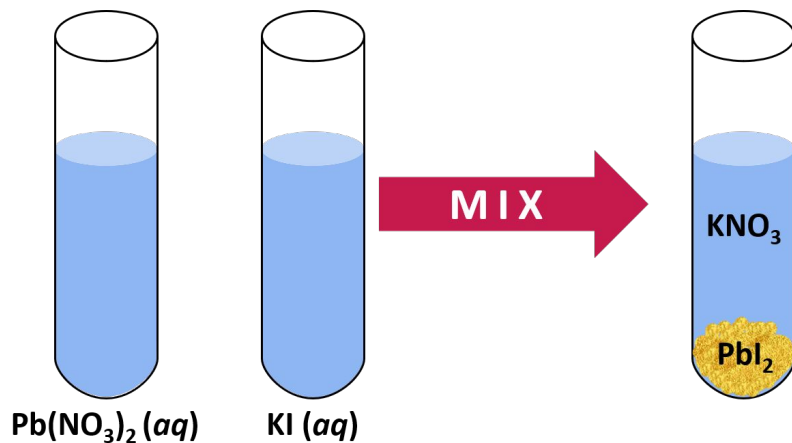


Make sure you can define the following terms:

- Soluble
- Aqueous
- Insoluble
 - How do we indicate in a chemical formula that it is aqueous or insoluble?
- Precipitate
- Precipitation (NOT the kind that falls from the sky !)

Precipitates

A precipitate is formed when two soluble salts/ionic compounds react in solution to form a new insoluble product



Predict the success of a double displacement reaction!



At a white board, determine if the reactions to the right will form a precipitate:

1. Write the two reactant FORMULAS
2. Determine the two possible products
3. Use the solubility table to determine the solubility of BOTH products

Reactants:

1. $\text{MgBr}_2(aq) + \text{AgNO}_3(aq)$
2. Silver nitrate + potassium phosphate
3. $\text{CuBr}_2 + \text{AlCl}_3$

Note: you don't need to balance for this activity

Photo of board work

Explain why there is no reaction if neither product forms a precipitate (what is happening at the level of ions in water before and after the reactant solutions mix).

What makes a reaction?

If BOTH products are soluble, has a reaction occurred?

Hint: on your board, draw a beaker of the **reactants** dissolved and a beaker of the **products** dissolved.

Part 3: Apply

Explain - Check in



Explain the difference between a precipitate that forms from a supersaturated solution and the precipitate that is formed in a double displacement reaction.

Diagrams are beneficial!

Practice Worksheet

Go to the folder and take one of the double displacement reaction practice worksheets.

Work independently to complete as many of the problems as necessary until you are confident. Be sure to check your answers as you go along.

Taking it Further: Generate your own double displacement reaction!

If you are given one (1) of the products, determine two potential reactants that would generate the product for both of the double displacement reactions on the right.

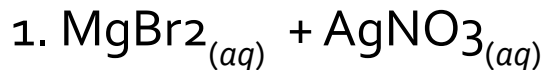
	Product
Reaction 1	$\text{Ag}_3\text{PO}_{4(s)}$
Reaction 2	$\text{CuCl}_{(s)}$

**Next slide(s) provide answers to
activities**

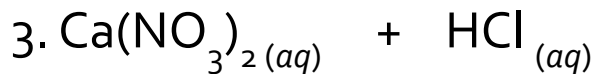
ANSWERS: USE IF YOU WERE AWAY Predict the success of a double displacement reaction!

PLEASE
CHECK
IN

Reactants:

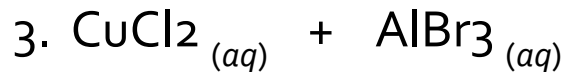
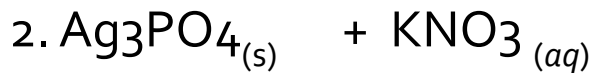
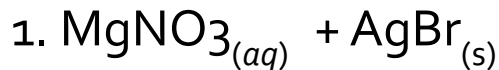


2. Silver nitrate +
potassium phosphate



Note: you don't need to
balance for this activity

Products:



If ONE product precipitates the reaction occurs. If BOTH products remained soluble, ions remain separated and there is NO reaction (NO new substance forms)